7 **(a)** State what is meant by an alternating current.

....

.... [1]

(b) A coil of wire is wound onto a hollow insulating tube to form a solenoid, as shown in Figure 7.1.

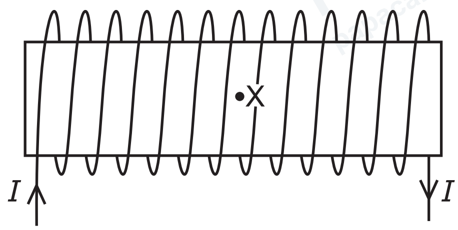

Figure 7.1

There is a constant current I in the solenoid in the direction indicated. There is a magnetic field inside and outside the solenoid due to this current.

(i) On Figure 7.1, draw five field lines to represent the magnetic field inside and outside the solenoid. [3]

(ii) On Figure 7.1, indicate, with the letters N and S, the magnetic poles of the solenoid. [1]

(c) The current I in the solenoid in **7(b)** now varies sinusoidally so that the flux density B of the magnetic field at point X varies with time t according to

$$B = 2.0 \cos 200\pi t$$

where B is in mT and t is in s.

(i) Show that the period of the alternating magnetic field is 0.010 s.

[1]

(ii) On Figure 7.2, sketch the variation of B with t between $t = 0$ and $t = 0.020$ s.

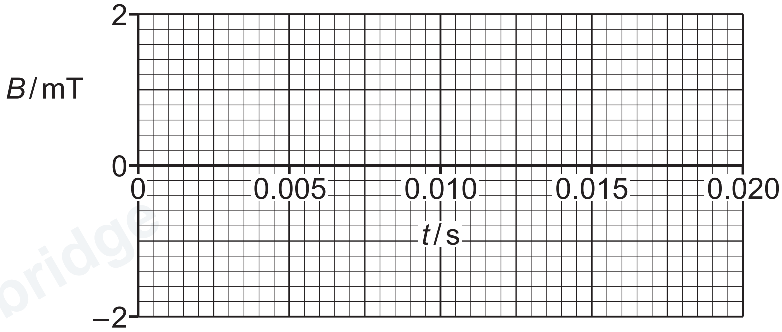

Figure 7.2

[2]

(d) A small coil is placed at point X inside the solenoid in **7(c)**, so that its axis is parallel to the axis of the solenoid.

The small coil has 45 turns, each with a radius of 2.8×10^{-3} m.

(i) Calculate the maximum magnetic flux Φ linked by the small coil. Give a unit with your answer.

Φ = ... unit .... [3]

(ii) Describe and explain how the variation of B with t in Figure 7.2 may be used to determine the maximum electromotive force (e.m.f.) induced in the small coil. No calculations are required.

....

....

....

....

.... [3]

[Total: 14]